ITAI 1378 · COMPUTER VISION — FINAL PROJECT

PPE Compliance Checker

Automated detection of hard hats, hi-vis vests, and safety goggles — flagging every worker as compliant or non-compliant.

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Tier 2 · mAP@50 0.842 on the held-out test split · 6-class detector · ~41 ms/image (T4)

GitHub: github.com/Tikskalani/ITAI-1378-Final_PPE-Compliance-Checker

The Problem

Preventable injuries + compliance liability on the job site

Workers who skip required PPE face some of the most common — and most preventable — worksite injuries: head trauma, struck-by incidents, and eye injuries.

Manual spot-checks don't scale. One safety officer cannot watch every worker continuously, and violations are logged inconsistently, if at all.

The result is both a safety gap and an OSHA-compliance liability.

1 **Who cares**

Safety officers / EHS managers, superintendents, construction & manufacturing firms, insurers

2 **Why it matters**

Head, eye & struck-by hazards are leading, largely preventable injuries

3 **Cost of failure**

Direct OSHA citations + injury and liability cost

The Solution

One sentence, one model

An image of a work area goes into a fine-tuned detector that locates each worker and each PPE item, then labels every person "Compliant" or "Non-compliant."

Job-site image → Detect worker + gear → Per-worker gear check → **Compliant / Non-compliant**

V2

Next step (V2): extend to a live or recorded video stream — track each worker across frames and log compliance events with timestamps, making it a deployable, portfolio-ready safety-monitoring tool.

Technical Approach

Object detection, done with the fastest reliable tool

1 **TECHNIQUE**

Multi-class object detection

Locate + identify people and several gear types in one pass.

2 **MODEL**

YOLO11 (Ultralytics), fine-tuned

Best accuracy-per-training-hour; trains on free Colab GPUs.

3 **FRAMEWORK**

PyTorch + Ultralytics

Standard, free, Colab-ready; built-in tracking for V2 video.

4 **COMPLIANCE LOGIC**

Rule-based, per worker

Associate gear to person; require detected hat + vest (goggles configurable).

Why YOLO11: top accuracy-per-training-hour for multi-class detection on a modest budget, and its inference speed is what carries the same model into the real-time V2 capstone without a rewrite.

Data Used

Public, already-labeled datasets — no manual collection

Roboflow Universe — PPE

Thousands of labeled images (a common set: ~19K / 3.6K / 1.9K)
person, hardhat, vest, goggles + NO- variants

SH17

8,099 images · ~75,994 instances
17 PPE + body-part classes (V2 scale-up)

Ultralytics Construction-PPE ← used for V1

1,416 images (1,132 train / 143 val / 141 test)
11 classes incl. helmet, vest, goggles, no_helmet, no_goggle

V1 uses Construction-PPE

- 1,416 images total
- 1,132 train / 143 val / 141 test
- Auto-downloads (178 MB), zero setup
- Standardize 640×640, light augmentation

Size: ample for a first fine-tune; scale to SH17 (8,099) for V2.

System Diagram

Input → inference → compliance decision

Job-site image
Image acquisition

→

YOLO11 detector
Model inference

→

Per-worker compliance logic
Associate gear → person

→

Box + Compliant / Non-compliant
Output & visualization

V2

Capstone: video stream → track workers across frames (ByteTrack) → timestamped compliance log.

Success Metrics

Target vs. measured — V3, trained natively on the six classes the rule uses

≥ 0.85	0.842	0.827
mAP@50 (target)	**mAP@50 — held-out TEST**	**mAP@50 — validation**
Detection quality across all classes	141 images never used for model selection	143 images · mAP@50-95 0.451

Supporting checks

- Test: precision 0.88 · recall 0.78 · mAP@50-95 0.455. V3 beats the V2 baseline on test by +0.018 mAP@50, and on mAP@50-95 on both splits.
- Latency: ~41 ms/image end-to-end at 960px on a T4 — the < 1 s/image target met 24× over.
- For reference, the all-11-class figure was 0.608 (V1) — capped by label defects, not model quality. Evidence on the next two slides.

Final result V3 final YOLO11s · 960px · 55 epochs · 0.83 h on a T4 · test 0.842 / val 0.827 · every evaluation image retained throughout.

How the Six Weeks Went

The plan vs. what actually happened

1	Set up Colab; downloaded and explored Construction-PPE	**Done — 1,416 imgs in YOLO format**
2	Fine-tuned YOLO11s — 40 epochs, ~17 min on a T4	**Done — first real mAP: 0.61**
3	Added per-worker compliance logic	**Done — end-to-end labels on images**
4	Evaluated on the held-out val set (143 imgs)	**0.61 vs 0.85 target — gap traced to rare no_* classes**
5	Built the demo — 6 annotated sample outputs	**Done — samples in results/**
6	Laptop lost — recovered all work from GitHub; docs and slides rebuilt	**Version control saved the project**

Challenges & Lessons Learned

What actually happened — and what saved us

CHALLENGE	OUTCOME
V2 (960px + oversampling) scored WORSE than V1 — 0.569 vs 0.608	Root cause: oversampling duplicated off-domain stock photos. Every no_* class fell; no_boots hit 0.000
The no_* labels contradict the gear labels	True no_boots is predicted as boots 50% of the time — irreducible error, not a training problem
The "none" class was quietly stealing real detections	It absorbed true vest 13%, gloves 8%, helmet 5% — removing it hands those back
Laptop lost near the deadline — all local files gone	Everything recovered from the public GitHub repo — commit early, commit often

System in Action — Sample Outputs

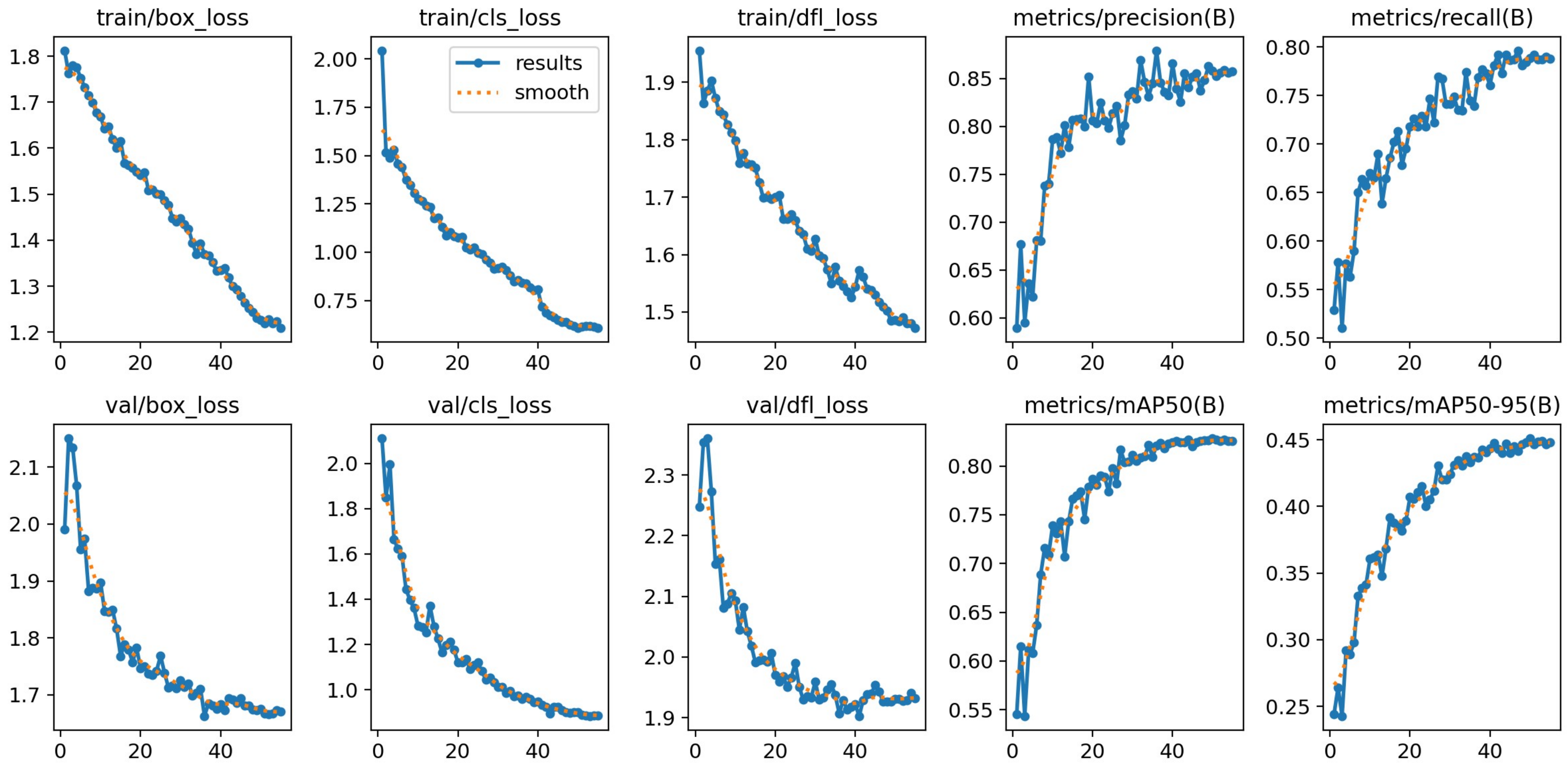
V3 outputs — positive-evidence rule, src/predict_compliance.py

How to read

- Green box = worker judged COMPLIANT
- Red box = NON-COMPLIANT, missing gear listed
- Rule: helmet + vest must be detected on the worker
- All 12 samples in results/v3_run/

Three Runs, One Diagnosis

V1 baseline → V2 regression → V3 fix. Every number measured.



The chain

- V1 11-class: 0.608 baseline
- V2 11-class: 0.569 — oversampling multiplied off-domain data
- Diagnosis: 5 classes have incoherent labels
- V3 6-class: test 0.842 · val 0.827

Per-Class Results

V3 kept classes (held-out test) vs the five classes dropped

Per-class mAP@50 — V3 kept, test split (left) vs dropped (right)

Person	0.836	none	0.505
vest	0.904	no_helmet	0.370
gloves	0.782	no_gloves	0.225
goggles	0.839	no_goggle	0.177
helmet	0.934	no_boots	0.000
boots	0.756		

Insight: V3 averages 0.842 across the six kept classes on the held-out test split — helmet 0.934 and vest 0.904, the two the compliance rule requires. The five dropped classes averaged 0.255 and are off-domain with self-contradictory labels. Every evaluation image was retained.